

University of Nottingham Malaysia

SCHOOL OF COMPUTER SCIENCE
A LEVEL 1 MODULE, AUTUMN SEMESTER 2021-2022

COMPUTER FUNDAMENTALS (COMP1027)

Time allowed: **ONE** Hour

An additional 30 minutes to cover IT issues and upload time

Answer ALL questions

Submit your answers containing all the work you wish to have marked as a **single PDF file, with each page in the correct orientation, to the appropriate Dropbox on the module's Moodle page.**

Use the standard naming convention for your document: [Student ID]_[Module Code]. Write your student ID number at the top of each page of your answers. Do not include your name.

Although you may use any notes or resources you wish to help you complete this open-book examination, the academic misconduct policy that applies to your coursework also applies here. You must be careful to avoid plagiarism, collusion or false authorship. This statement refers to, and does not replace, the University policy which stipulates severe penalties for academic misconduct.

Staff are not permitted to answer assessment or teaching queries during the period in which your open-book examination is live. If you spot what you think may be an error on the exam paper, note this in your submission but answer the question as written.

IMPORTANT: Lateness penalties will not be applied, and late submission will not be accepted.

Marks available for sections of questions are shown in brackets in the right-hand margin

No calculators are permitted in this examination

Allowed time

60

60

99.99 %

100

100

			3	9	9	5	10	2	100				100
Q	Topic	Sub	Taxonomy			Difficulty Level			Estimated time	%time	Weights	Weight/Sub	
K C A L M H													
1	Number Representation and Binary Arithmetic										18		
		a			1		1		3	4.9995		6	
		b			1		1		4	6.666		6	
		c (i)			1		1		2	3.333		2	
		c (ii)			1		1		2	3.333		2	
		c (iii)			1		1		2	3.333		2	
2	Boolean Algebra and Digital Logic										30		
		a		1	1		1		5	8.3325		15	
		b (i)		1		1			1	1.6665		3	
		b (ii)		1	1			1	4	6.666		6	
		b (iii)		1	1		1		3	4.9995		6	
3	Adders, MUX and ALU										27		
		a (i)		1		1			2	3.333		3	
		a (ii)		1		1			2	3.333		3	
		b	1			1			2	3.333		6	
		c		1	1			1	6	9.999		15	
4	CPU Instructions and Networks										25		
		a	1			1			3	4.9995		4	
		b (i)	1				1		5	8.3325		14	
		b (ii)		1			1		2	3.333		4	
		c		1			1		2	3.333		3	
	Reading instructions & all								1	1.6665			
	Revise								9	14.9985			

1. Number Representation and Binary Arithmetic [18%]

- a) Given Table 1, calculate the 6 values labelled *a-f*. Your representations for Binary, 1's Complement and 2's Complement must be in 12-bits width. (6)

Table 1.

Decimal	Binary	1's Complement	2's Complement
1010	<i>a</i>	<i>b</i>	<i>c</i>
<i>f</i>	<i>e</i>	<i>d</i>	1011 1011 1010

[A]

correct answer (1 mark)

a =

1010 / 2	0
505 / 2	1
252 / 2	0
126 / 2	0
63 / 2	1
31 / 2	1
15 / 2	1
7 / 2	1
3 / 2	1
1 / 2	1
0 / 2	0
0 / 2	0

Remainders

a = **0011 1111 0010** (1 mark)b = **1100 0000 1101** (inverts all bits) – (1 mark)

c = 1100 0000 1101
 1 + (adds "1")
 = **1100 0000 1110** (1 mark)

d = 1011 1011 1010
 1 – (minus "1" to revert to 1's complement)
1011 1011 1001 (1 mark)

e = 1011 1011 1001 (inverts all bit to revert to binary)
 = **0100 0100 0110** (1 mark)

f =

2^{11}	2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
0	1	0	0	0	1	0	0	0	1	1	0
	1024				64				4	2	

= 1024 + 64 + 4 + 2
 = **- 1094** (1 mark)

P.T.O

b) Given the following values of x, y and z below (in Decimal numbers):

- $x = -1027$,
- $y = 540$,
- $z = -43$

Calculate, showing your working in details, the binary representation in 12-bit TWO's complement. (6)

(A)

Workings (1 mark), correct answer (1 mark)

x =

1027 / 2	1
513 / 2	1
256 / 2	0
128 / 2	0
64 / 2	0
32 / 2	0
16 / 2	0
8 / 2	0
4 / 2	0
2 / 2	0
1 / 2	1
0 / 2	0

(1 mark)

= 0100 0000 0011
 = 1011 1111 1100 (inverts all bits)
 = 1 + (adds "1")
 = **1011 1111 1101** (1 mark)

y =

540 / 2	0
270 / 2	0
135 / 2	1
67 / 2	1
33 / 2	1
16 / 2	0
8 / 2	0
4 / 2	0
2 / 2	0
1 / 2	1
0 / 2	0
0 / 2	0

(1 mark)

= **0010 0001 1100** (1 mark)

P.T.O

z =	43 / 2	1
	21 / 2	1
	10 / 2	0
	5 / 2	1
	2 / 2	0
	1 / 2	1
	0 / 2	0
	0 / 2	0
	0 / 2	0
	0 / 2	0
	0 / 2	0
	0 / 2	0

(1 mark)

= 0000 0010 1011
 = 1111 1101 0100 (inverts all bits)
 = 1 + (adds "1")
 = **1111 1101 0101** (1 mark)

- c) Using your **12-bits TWO's complement** answers in (b), calculate the following the sums. Show your working in detail. Your answer should be in 12-bits width. (6)
- $x + y$
 - $y - z$
 - $x + z$

{ A }

Workings (1 mark), correct answer (1 mark)

$$\begin{array}{rcl}
 x + y = & \begin{array}{r} x = 1011\ 1111\ 1101 \\ y = 0010\ 0001\ 1100 \\ \hline \end{array} & + \begin{array}{l} (1\ \text{mark}) \\ (1\ \text{mark}) \end{array} \\
 & \underline{\mathbf{1110\ 0001\ 1001}} &
 \end{array}$$

$$y - z = y + (-z)$$

$$\begin{array}{rcl}
 y = & 0010\ 0001\ 1100 & \\
 -z = & 1111\ 1101\ 0101 & \\
 \hline & \mathbf{1\ 0001\ 1111\ 0001} & + \begin{array}{l} (1\ \text{mark}) \\ (1\ \text{mark}) \end{array}
 \end{array}$$

(in 2's complement ignore the **carry out**)

$$\begin{array}{rcl}
 x + z = & \begin{array}{r} x = 1011\ 1111\ 1101 \\ z = 1111\ 1101\ 0101 \\ \hline \end{array} & + \begin{array}{l} (1\ \text{mark}) \\ (1\ \text{mark}) \end{array} \\
 & \underline{\mathbf{1\ 1011\ 1101\ 0010}} &
 \end{array}$$

(in 2's complement ignore the **carry out**)

P.T.O

2. Boolean Algebra & Digital Logic [30%]

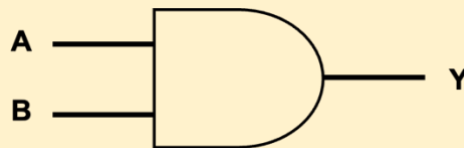
- a) In the lectures, you have been introduced to **three** basic & **two** universal gates. List them by drawing their respective logic diagram and truth table with A, B as inputs and Y as the output. Please label the inputs and output in your logic diagram. (15)

{ C, A }

Logic diagram with correct labelling (2 mark), correct truth table (1 mark)

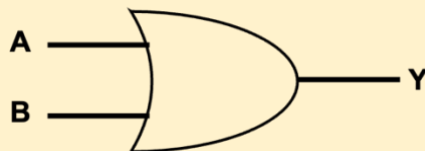
Basic Gates

AND – (3 marks)



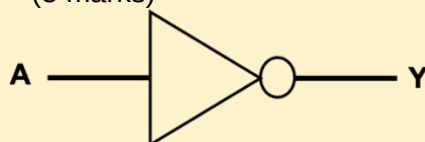
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR – (3 marks)



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT – (3 marks)

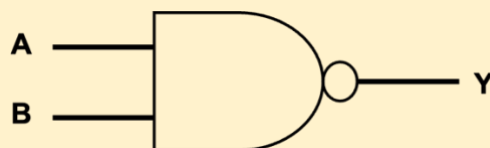


A	Y
0	1
1	0

Input can be either A or B

Universal Gates

NAND – (3 marks)



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

NOR – (3 marks)



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

P.T.O

- b) Given the below **truth table** in Table 2:

Table 2.

X	Y	Z	F
1	1	1	1
1	1	0	0
1	0	1	1
1	0	0	0
0	1	1	1
0	1	0	0
0	0	1	0
0	0	0	0

- i. Write out the expression (pre-simplified)

(3)

{C}

$$F(X,Y,Z) = XYZ + XY'Z + X'YZ \text{ (3 marks)}$$

Identity Name	AND Form	OR Form
Identity Law	$1x = x$	$0+x = x$
Null (or Dominance) Law	$0x = 0$	$1+x = 1$
Idempotent Law	$xx = x$	$x+x = x$
Inverse Law	$x\bar{x} = 0$	$x+\bar{x} = 1$
Commutative Law	$xy = yx$	$x+y = y+x$
Associative Law	$(xy)z = x(yz)$	$(x+y)+z = x+(y+z)$
Distributive Law	$x+yz = (x+y)(x+z)$	$x(y+z) = xy+xz$
Absorption Law	$x(x+y) = x$	$x+xy = x$
DeMorgan's Law	$(\overline{xy}) = \bar{x}+\bar{y}$	$(\overline{x+y}) = \bar{x}\bar{y}$
Double Complement Law	$\overline{\bar{x}} = x$	

Extracted from: L. Null, L. and J. Lobur, *The essentials of computer organisation and architecture*, Jones & Bartlett Publishers, 2003 (Fig. 1.3, P.26)

Figure 1.

- ii. Using the given Basic Identities of Boolean Algebra in Figure 1, simplify your expression in b(i). Detail each step. (6)

{C, A}

$$XYZ + XY'Z + X'YZ$$

$$XZ(Y + Y') + X'YZ \text{ (Distributive Law) – (1 marks)}$$

$$XZ.1 + X'YZ \text{ (Complement Law) – (1 marks)}$$

$$XZ + X'YZ \text{ (Identity Law) (1 marks)}$$

$$Z(X'Y + X) \text{ (Distributive Law) (1 marks)}$$

$$Z(Y + X) \text{ (Absorption Law) (1 marks)}$$

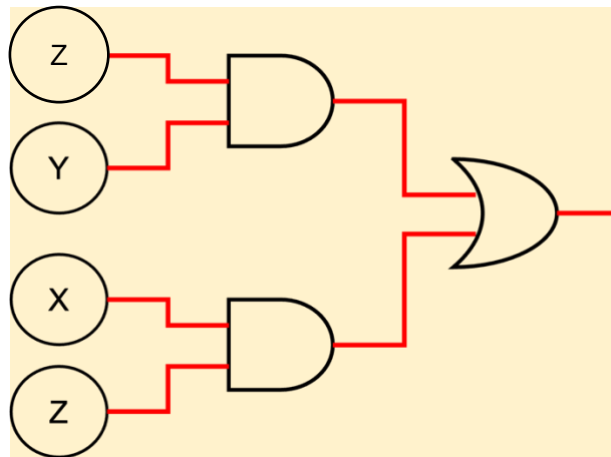
$$ZY + ZX \text{ (1 marks for correct answer)}$$

- iii. Based on your answer in (b), draw the Logic Circuit Diagram to implement the circuit (using the relevant combinations of ONLY basic gates). (6)

{C, A}

Correct basic gates used – 2 AND (4 marks) & 1 OR (2 marks)

P.T.O



3. Adders, MUX and ALU [27%]

a. Explain the **significant** difference between:

i. Combinatorial Logic and Sequential Logic (3)

[C]

Combinatorial logic consists of only gates dependable on the inputs at the same instant of time. In comparisons, sequential logic consists of logic gates and memory elements whereby outputs are not only dependent on the present input but also on previous outputs (3 marks)

ii. Multiplexer (MUX) and De-multiplexer (DEMUX) (3)

[C]

Multiplexer (data selector) takes in multiple inputs, select on one input and forward it to the output. In comparisons, demultiplexer (data distributor) takes in one input and distribute or deliver it out via multiple outputs (3 marks)

b. List down the differences that sets a Half Adder and Full Adder apart in terms of functionality and gates utilised. (6)

[K]

	Half Adder	Full Adder
Functionality (3 marks)	Adds two 1-bit digits. Produces a sum of the two inputs Have a carry output	Performs an addition operation on three one-bit binary numbers. Produces a sum of the three inputs and carry value
Gates utilised (3 marks)	Consists of one EX-OR & one AND gate	Consists of two EX-OR, two AND & one OR gate

c. For the ALU design (given in Figure 2), show (with *full working details*) how this ALU can achieve each of the listed output functionalities (*out*) according to the given corresponding control values in Table 3 (on page 4).

In your working details, you need to state what each of the variables (x_1 , x_2 , y_1 , y_2 , K_1 , K_2 , Out_1 , Out) represents/contains in each case (a, b, c; in the table below) (15)

P.T.O

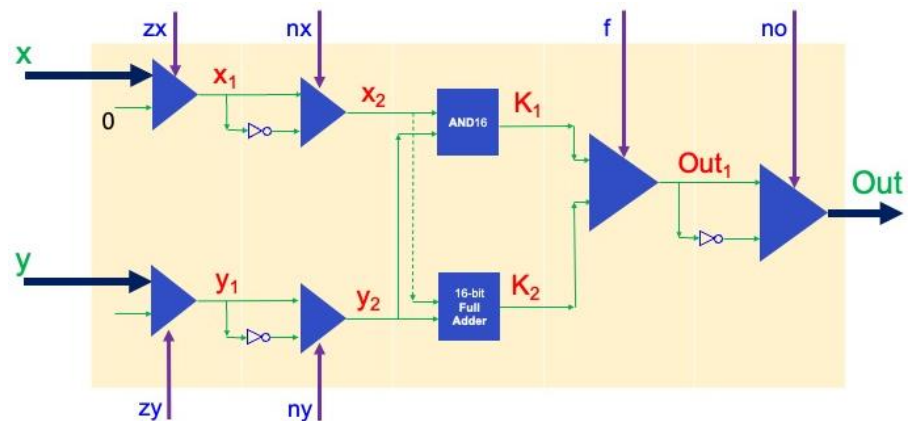


Figure 2.

Table 3.

	zx	nx	zy	ny	f	no	Out
a.	0	0	1	1	0	1	!x
b.	1	1	0	1	1	1	y + 1
c.	0	0	0	1	1	1	y - x

Hint: $-A = A \sim + 1$ { where $A \sim$ is the bit-wise Not(A) }.

	x_1	x_2	y_1	y_2	k_1	k_2	out_1	out
a.	x	x	0	1	$= x.1$ $= x$	$= x + 1$	!x	!x
b.	0	1	y	$= !y$ $= (-y-1)$	$= (1).(-y-1)$ $= -y - 1$	$= 1 + (-y - 1)$ $= 1 - y - 1$ $= -y$	Since $-A = A \sim + 1$ Therefore: $(-A) \sim = A + 1$ So, $(-y) \sim = y + 1$	y + 1
c.	x	x	y	$= !y$ $= (-y-1)$	$= (x).(-y-1)$ $= -xy - x$	$= x + (-y-1)$ $= x - y - 1$	$(x - y - 1) \sim$ $= -(x - y - 1) - 1$ $= -x + y + 1 - 1$ $= -x + y$ $= y - x$	y - x

Correct representations:

- a. $x_2 = 1$ mark; $y_2 = 1$ mark; $k_1 = 1$ mark; $k_2 = 1$ mark; $out_1 = 1$ mark (total = 5 marks)
- b. $x_2 = 1$ mark; $y_2 = 1$ mark; $k_1 = 1$ mark; $k_2 = 1$ mark; $out_1 = 1$ mark (total = 5 marks)
- c. $x_2 = 1$ mark; $y_2 = 1$ mark; $k_1 = 1$ mark; $k_2 = 1$ mark; $out_1 = 1$ mark (total = 5 marks)

P.T.O

4. CPU Instructions and Networks [25%]

- a. Based on your understanding of the Hack Computer, describe the **Fetch-Execute Cycle** process for A- or C- Instructions. (4)

{K}

Fetch: Access the instruction (to be executed) in Instructions Memory

- Address in PC
- Access ROM[PC] to read instruction

(2 marks)

Execute: Decode A- or C- instruction for execution

- If C-instruction, then decode the 3 parts (**dest, comp, jump**) prior to executing

(2 marks)

Table 4.

	Layer Name	Provide/Facilitate
7	-	
6	-	
5	-	
4	-	
3	-	
2	-	
1	-	

- b. Table 4 above shows the OSI Network stack:
- i. Give the names for layers 1 till 7 and list what each layer provide/facilitate. (14)

{K}

	Layer Name	Provide/Facilitate	
7	Application	Interface for applications to access network	(2 marks)
6	Presentation	Enables common language/format	(2 marks)
5	Session	Establish & maintain good communication between clients	(2 marks)
4	Transport	Break into segments, handle end-end errors	(2 marks)
3	Network	Routing & packet delivery through reliable connections	(2 marks)
2	Data Link	Prepare data frames for physical transmission	(2 marks)
1	Physical	Transmit bits over the network hardware	(2 marks)

- ii. Give **one** example of common protocols for layers two, three, four and seven. (4)

{C}

Layer two – **Ethernet** (1 mark)

Layer three – **IPv4/IPv6** (1 mark)

Layer four – **TCP/UDP** (1 mark)

Layer seven – **HTTP/FTP/DNS/POP3/IMAP/TELNET** (1 mark)

- c. Describe the **main difference** between and IPv4 address and an IPv6 address and why it is necessary to move from IPv4 to IPv6. (3)

{C}

IPv4 is a 32-bit IP address that uses numeric addressing whereas IPv6 is a 128-bit IP address that uses an alphanumeric addressing method. Given that IPv4 address space are quickly being used up and running out. For this reason, it is necessary to move from IPv4 to IPv6 (3 marks)

END